

Bindu

An Image Processing Workspace

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The Big Picture

- ▶ An image is a 2D signal: analyze it, transform it, compact it.
- ▶ Every tool here is one of those three moves.

Analysis

- Color channel separation (R, G, B)
- Per-channel and combined histograms
- Luminance histogram
- 2D FFT log-magnitude spectra
- Color space comparison (RGB vs. YCbCr)

Processing

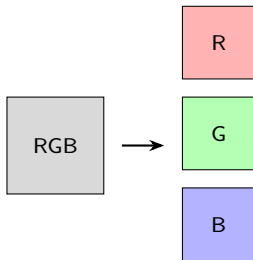
- Gaussian and box blur
- Laplacian sharpening
- Unsharp masking
- Per-channel or merged filtering
- White scratch / dot removal

Compression

- Partial image reconstruction from channel spectra
- Fourier energy pruning (lossy)
- PSNR & MSE quality metrics

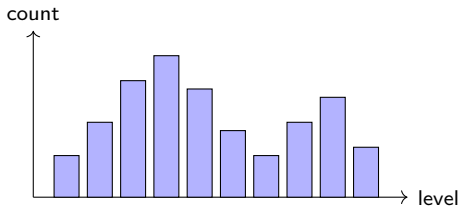
Channel Separation

- ▶ $RGB = \text{three stacked signals } R(x, y), G(x, y), B(x, y).$
- ▶ BGR vs. RGB load order is a classic bug, caught by viewing channels.



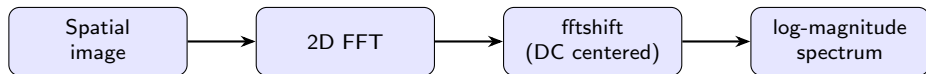
Histograms: Statistics of a Signal

- ▶ Histogram = empirical distribution of pixel intensities.
- ▶ Luminance blends channels *per pixel first*: $Y = 0.299R + 0.587G + 0.114B$.



Frequency Content: The 2D FFT

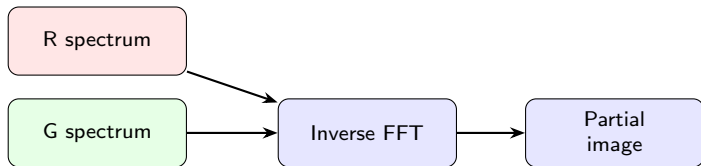
- ▶ Image $\rightarrow$ sum of 2D sinusoids of varying spatial frequency.
- ▶ `fftshift` centers DC; log-magnitude makes the spectrum viewable.



Parseval: energy in space = energy in frequency

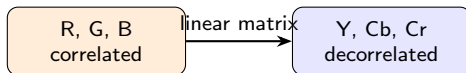
Partial Reconstruction

- ▶ Inverse FFT is linear: rebuild from a subset of channel spectra.
- ▶ *Phase*, not magnitude, carries most of the structure.



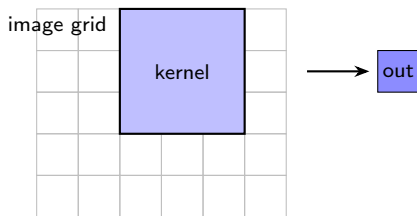
Color Space: RGB vs. YCbCr

- ▶ Linear change of basis: decorrelates brightness from color.
- ▶ Y carries detail; C_b, C_r are smooth $\Rightarrow$ energy compaction.



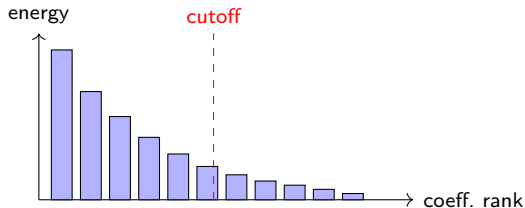
Spatial Filtering: Convolution

- ▶ A kernel slides over the image; each pixel becomes a weighted local sum.
- ▶ Low-pass = blur (box, Gaussian); high-pass = sharpen (Laplacian, unsharp).



Lossy Compression: Fourier Energy Pruning

- ▶ Rank frequency coefficients by energy; keep a target energy %.
- ▶ Real images: conjugate-symmetric spectrum $\Rightarrow$ store half.



Lossless Compression: Integer Wavelet Lifting

- ▶ Integer Haar wavelets separate the image into averages and details.
- ▶ Keep every coefficient and compress the resulting patterns.
- ▶ Reverse the steps to recover the exact RGB image.

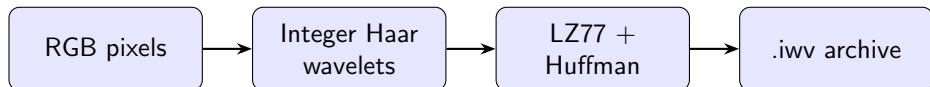


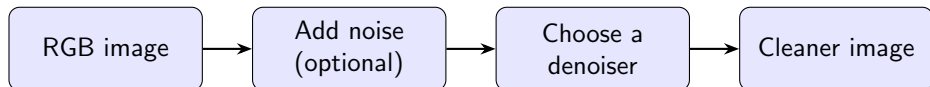
Photo Restoration

- ▶ White top-hat subtracts the opened grayscale image to reveal bright marks.
- ▶ A repair mask selects the damaged areas.
- ▶ Inpainting estimates replacement colors from nearby pixels.



Noise: Models and Denoising

- ▶ Gaussian noise adds random variation; salt-and-pepper adds black or white dots.
- ▶ Median, Gaussian blur, or non-local means reduce noise.
- ▶ Denoising estimates a cleaner image and may soften some details.



Recurring Threads

